

Numerical Implementation and Reproducibility

A Nonlocal Attraction–Repulsion Transport Equation

Companion to `matlabcode.m` (version of September 2026)

This document describes the supplied `matlabcode.m` script, in the order of the numerical examples in the manuscript: one dimension followed by two dimensions, and then the refinement study. Each stationary calculation starts with the mathematical problem, explains the representation being used, and then describes its discretization and output. The numerical target is the stationary measure selected by the manuscript’s characterization, rather than the background measure itself. Equation, proposition, table and figure numbers refer to the manuscript: Proposition 7.1 with the finite Riesz equation (7.2), the Green-kernel reconstruction (7.3) and the zero-flux (force-balance) condition (7.4); Proposition 6.4 with the general reconstruction (6.19) and Boggio’s formula (6.20); Proposition 7.3 with the disk formula (7.11)–(7.12) and the force verification (7.13); Table 1 for the constants; Figures 1 and 2 for the two experiments.

How the four stationary calculations fit together. The same scheme is followed in both dimensions. First the zero-flux condition itself is solved: in one dimension the force balance (7.4) on $I = [-L, L]$ (Section 2.1), in two dimensions the radial force balance (7.13) on the disk B_R (Section 3.1). In both cases the support parameter (L , respectively R) is prescribed, the nonnegative stationary measure of mass one is obtained by nonnegative least squares at test points, and the support parameter is then selected as the minimizer of the root-mean-square force residual; the right panel of each figure shows this residual as a function of the prescribed support parameter. Second, the Green-kernel formula of the corresponding proposition is evaluated independently on the same support: the interval formula (7.3) (Section 2.2) and the disk formula (7.11)–(7.12) (Section 3.2), both with all constants fixed analytically and without any fitted coefficient. Finally, a long-time particle solution supplies a third distribution to compare with the stationary computations. Section 4 reports how each computation changes under refinement of its discretization parameters and of the terminal time.

1 Running the code

Place `matlabcode.m` in a writable working directory and run `matlabcode`. The script clears workspace variables, closes open figures and clears the command window; it contains its own local functions and needs no external input data or toolbox (it calls `lsqnonneg`, `fzero`, `fminbnd`, `besselj`, `betainc`, `gamma`, `beta`, `quantile`, `histcounts`, `xline`, `yline`, `sgtitle`; a projected-gradient fallback replaces `lsqnonneg` if it is missing). The flag `runRefinementStudy` at the top of the script (default `true`) controls Part III. Without the refinement study the script produces the two figures of the manuscript in about one minute; the refinement study adds roughly fifteen minutes, most of it in the finest zero-flux solve and in the particle systems with the largest number of particles or the longest terminal time.

The generated files are `fig_1d_a_greater_r_phi_compare.pdf` and `fig_2d_ar_phi_compare.pdf` (Figures 1 and 2 of the manuscript, with PNG counterparts) and, with the refinement study, `fig_refinement_study.pdf`, the text log `refinement_results.txt`,

the L^AT_EX table macros `refinement_tables.tex` that are included in Section 4 of this document, and `refinement_results.mat`. Export uses `exportgraphics` with a `print` fallback; existing outputs are overwritten. Console output includes the selected support parameters, residuals, masses, CDF discrepancies and the diagnostics of the Green reconstruction. The numbers quoted in this document are those printed by the script (run under GNU Octave 8.4 with the supplied plotting shims; the stationary computations are deterministic, and the particle results are independent of the random initial perturbation because the particle systems have equilibrated by $T = 50$, see Section 2.3).

2 One-dimensional experiment

2.1 The zero-flux stationary state on an interval (Proposition 7.1, condition (7.4))

Original problem. The target is a nonnegative probability density ϕ_I supported on an unknown symmetric interval $I = [-L, L]$ with zero force on its interior,

$$V'(x) = (1+r) \int_{-L}^L \operatorname{sgn}(x-y) |x-y|^r \phi_I(y) dy, \quad x \in (-L, L), \quad \phi_I \geq 0, \quad \int_{-L}^L \phi_I(y) dy = 1, \quad (1)$$

with $V = \psi_a * \omega_m$. This is the force-balance (zero-flux) condition (7.4) of Proposition 7.1, imposed at every interior point rather than at one point x_* only; for the even background considered here the force is odd, so imposing (1) for $x \in (0, L)$ is equivalent to imposing it on all of $(-L, L)$. Both the profile and the half-width are unknown. Differentiating (1) gives the finite Riesz equation (7.2), $r(1+r) \int_I |x-y|^{r-1} \phi_I(y) dy = V''(x)$; the script solves (1) directly and uses (7.2) only as a cross-check.

Why a least-squares formulation with a prescribed L ? For a prescribed L , equation (1) is linear in ϕ_I , and the mass and sign conditions are a linear constraint and a cone constraint. Nonnegative least squares over more test points than unknowns imposes the sign constraint exactly, treats the force equation and the mass equation together, and returns a sampled force residual. For the correct half-width all conditions can be met simultaneously and the residual is at the level of the discretization error; for a wrong half-width they cannot, and the residual grows. The half-width is therefore selected as the minimizer of the residual. The same construction is used in two dimensions (Section 3.1), so that the two experiments test the stationary characterization in the same way.

Numerical calculation. The background and exponents are $\omega_m = \frac{m}{2} \mathbf{1}_{[-1,1]}$, $(a, r, m) = (0.8, 0.2, 1.8)$, so that $V'(x) = \frac{m}{2} (|x+1|^{1+a} - |x-1|^{1+a})$ in closed form. For a prescribed L the interval is partitioned into $n = 320$ uniform cells $[e_{j-1}, e_j]$ of width $h = 2L/n$ and $\phi_h = \sum_j c_j \mathbf{1}_{[e_{j-1}, e_j]}$; since ϕ_I is even, the $n/2$ values on the right half are the unknowns, the mirrored cells being paired. The force is tested at $n_{\text{test}} = 240$ midpoints x_i of a uniform grid on $(0, L)$. Every cell integral of the odd kernel is exact,

$$\int_{\ell}^u \operatorname{sgn}(x-y) |x-y|^r dy = \frac{1}{1+r} \begin{cases} (x-\ell)^{1+r} - (x-u)^{1+r}, & x \geq u, \\ -(u-x)^{1+r} + (\ell-x)^{1+r}, & x \leq \ell, \\ (x-\ell)^{1+r} - (u-x)^{1+r}, & \ell < x < u, \end{cases} \quad (2)$$

so the discrete force system is $Ac = b$ with $A_{ij} = (1+r) \int_{e_{j-1}}^{e_j} \operatorname{sgn}(x_i - y) |x_i - y|^r dy$ (mirrored columns added) and $b_i = V'(x_i)$. The code solves

$$\min_{c \geq 0} \|Ac - b\|_2^2 + w^2 (h \sum_j c_j - 1)^2, \quad w = 120,$$

with `lsqnonneg` (default settings), clips and normalizes the result to mass one, and defines the force residual $\text{Res} = Ac - b$ at the test points. The half-width is selected in two stages: a scan of the RMS residual $\|\text{Res}\|_2/\sqrt{n_{\text{test}}}$ over $L \in \{0.240, 0.2425, \dots, 0.310\}$ (Table 1), then `fminbnd` on the bracket formed by the two scan points adjacent to the minimum (tolerance 10^{-7}). This gives

$$L = 0.2726852, \quad \|\text{Res}\|_2/\sqrt{240} = 2.5 \times 10^{-7}, \quad \|\text{Res}\|_\infty = 2.4 \times 10^{-6}.$$

The residual is V -shaped in L (right panel of Figure 1, logarithmic scale): it decreases by four orders of magnitude towards the minimizer and increases again beyond it. The CDF F_I is obtained by integrating ϕ_h over the cells and dividing by its final mass.

Cross-check with the differentiated form. The script also solves the finite Riesz equation (7.2) by midpoint collocation on the same n cells, with the diagonal cells integrated exactly, and selects L there from the mass condition with `fzero`. The two half-widths differ by 2.9×10^{-5} at $n = 320$ and the two densities by 6×10^{-4} (relative to the maximum, over the central 90% of the cells); the difference decreases under refinement (Table 3). The two discretizations of the same stationary state are thus consistent, and the direct formulation (1) is the one displayed.

Meaning of the output. The result is the stationary density ϕ_I , its cumulative mass F_I , and the half-width selected by the zero-flux condition; the residual measures how well the sampled force balance is satisfied after the mass normalization. It does not by itself quantify the discretization error, which is estimated in Section 4.2. These outputs form the first density and CDF curves and the right panel of Figure 1.

2.2 Green/Boggio reconstruction of the interval profile (Proposition 7.1, formula (7.3))

Original problem. The target is the second stationary curve in Figure 1, the Green-kernel reconstruction (7.3) of Proposition 7.1, which the proposition states as an explicit formula for the same measure ϕ_I that Section 2.1 computes from the zero-flux condition. For $I = [-L, L]$, $\Omega = \mathbb{R} \setminus I$ and $\sigma = r/2$, the proposition gives

$$\phi_I = (Q_{a,r} - c_{1,r}(-\Delta)^\sigma H)|_I, \quad H(x) = c_{1,r}^{-1} \int_\Omega G_\Omega^\sigma(x, y) Q_{a,r}(y) dy \quad (x \in \Omega), \quad H = 0 \text{ on } I, \quad (3)$$

with the exterior interval Green kernel

$$G_\Omega^\sigma(x, y) = \kappa_{1,\sigma} |x-y|^{r-1} \int_0^{\eta_I(x,y)} \frac{t^{\sigma-1}}{\sqrt{1+t}} dt, \quad \eta_I(x, y) = \frac{(x^2 - L^2)(y^2 - L^2)}{L^2 |x-y|^2}, \quad \kappa_{1,\sigma} = \frac{1}{4^\sigma \Gamma(\sigma)^2}, \quad (4)$$

which is the $d = 1$ exterior form of Boggio's formula (6.20) obtained from the interior formula by the Kelvin transform. Here $Q_{a,r} = \mathcal{L}_r V$ is the source of the free-boundary problem and $c_{1,r} = \gamma_{1,r}^{-1}$ the normalization of $\mathcal{L}_r = c_{1,r}(-\Delta)^{1+\sigma}$ (Table 1).

Three analytic reductions. The implementation evaluates (3) literally, after three exact simplifications.

(i) *The constant $c_{1,r}$ cancels.* Because H vanishes on I , for $x \in I$ the singular-integral form of the fractional Laplacian contains no principal value:

$$(-\Delta)^\sigma H(x) = C_{1,\sigma} \int_{\mathbb{R}} \frac{H(x) - H(y)}{|x-y|^{1+2\sigma}} dy = -C_{1,\sigma} \int_\Omega \frac{H(y)}{|x-y|^{1+2\sigma}} dy, \quad C_{1,\sigma} = \frac{4^\sigma \Gamma(\frac{1}{2} + \sigma)}{\sqrt{\pi} |\Gamma(-\sigma)|}, \quad (5)$$

where $C_{1,\sigma}$ is the constant that makes (5) agree with the Fourier multiplier $|\xi|^{2\sigma}$ used in the manuscript. Writing $W := c_{1,r}H = \int_{\Omega} G_{\Omega}^{\sigma}(\cdot, z)Q_{a,r}(z) dz$, the reconstruction becomes

$$\phi_I(x) = Q_{a,r}(x) + C_{1,\sigma} \int_{\Omega} \frac{W(y)}{|x-y|^{1+2\sigma}} dy, \quad x \in I. \quad (6)$$

The only constants that remain are $\kappa_{1,\sigma}$ inside G_{Ω}^{σ} , $C_{1,\sigma}$, and the constant in $Q_{a,r}$. The latter multiplies both terms of (6), so it only fixes the total mass; the *relative* weight of the source and of the nonlocal correction is fixed by $\kappa_{1,\sigma}C_{1,\sigma}$ and is not adjustable. This is the difference from the two-dimensional routine of Section 3.2, where a relative coefficient is fitted.

(ii) *Closed form of the source.* By (6.10), $Q_{a,r} = C_{1,a,r} \omega_m * |x|^{-(1-\beta)}$ with $\beta = a - r = 0.6$ and $C_{1,a,r}$ from Table 1. For the uniform background the convolution is elementary:

$$Q_{a,r}(y) = c_Q [\operatorname{sgn}(y+1)|y+1|^{\beta} - \operatorname{sgn}(y-1)|y-1|^{\beta}], \quad c_Q = \frac{C_{1,a,r} m}{2\beta}, \quad (7)$$

$$C_{1,a,r} = \frac{\Gamma(\frac{1-\beta}{2})\Gamma(\frac{2+a}{2})\Gamma(-\frac{1+r}{2})}{\sqrt{\pi}\Gamma(\frac{\beta}{2})\Gamma(-\frac{1+a}{2})\Gamma(\frac{2+r}{2})}.$$

$Q_{a,r}$ is positive and even, has a cusp $|y \mp 1|^{\beta}$ at $y = \pm 1$ and decays like $2\beta c_Q y^{\beta-1}$. The script checks (7) against direct quadrature of (6.10) (agreement to 10^{-10}), and for $|y| > 2$ evaluates the bracket in the cancellation-free form $|y|^{\beta}[\expm1(\beta \log(1+1/|y|)) - \expm1(\beta \log(1-1/|y|))]$, which is needed because the tail quadrature of Step 1 below reaches $|y| \sim 10^{18}$. With the present parameters $c_Q = 0.42361$ and $C_{1,a,r} = 0.28241$.

(iii) *Boggio's integral is an incomplete beta function.* The substitution $u = t/(1+t)$ gives

$$\int_0^{\eta} \frac{t^{\sigma-1}}{\sqrt{1+t}} dt = B\left(\frac{\eta}{1+\eta}; \sigma, \frac{1}{2} - \sigma\right) = B(\sigma, \frac{1}{2} - \sigma) I_{\eta/(1+\eta)}(\sigma, \frac{1}{2} - \sigma), \quad (8)$$

where $I_x(p, q)$ is the regularized incomplete beta function (`betainc`). The same identity with $(1+t)^{-1}$ in place of $(1+t)^{-1/2}$, i.e. with parameters $(s, 1-s)$, is used in two dimensions. The argument is formed as $\eta/(1+\eta) = N/(N+L^2|x-y|^2)$ with $N = (x^2 - L^2)(y^2 - L^2)$, which is exact on the diagonal $x = y$ ($\eta = \infty$, value $B(\sigma, \frac{1}{2} - \sigma) = 11.9058$) and never leaves $[0, 1]$. Numerically $\kappa_{1,\sigma} = 9.6188 \times 10^{-3}$ and $C_{1,\sigma} = 9.0314 \times 10^{-2}$.

Behaviour at infinity. The source decays only like $y^{\beta-1} = y^{-0.4}$ and $G_{\Omega}^{\sigma}(y, z) \sim \kappa_{1,\sigma}|z|^{r-1}B(\sigma, \frac{1}{2} - \sigma)$ for $|z| \rightarrow \infty$, so W inherits the slow decay $W(y) \sim Ay^{a-1} = Ay^{-0.2}$, with

$$A = \kappa_{1,\sigma}B(\sigma, \frac{1}{2} - \sigma)2\beta c_Q[B(\beta, r) + B(1-a, r) + B(\beta, 1-a)] \quad (9)$$

obtained by scaling the Green representation. Consequently both exterior integrals in (6) converge slowly: truncating the z -integral of W at $Y_{\max}$ omits $O(Y_{\max}^{a-1})$, and truncating the y -integral of (6) omits $O(Y_{\max}^{-(1-\beta)}) = O(Y_{\max}^{-0.4})$, essentially an x -independent constant added to the density. Table 5 shows that without these tails the reconstructed mass is 0.94 even at $Y_{\max} = 10^4$; the implementation therefore integrates the tails explicitly, as described next.

Numerical calculation. The half-width is the L selected in Section 2.1 (no rounding) and the $n = 320$ interior cells and midpoints x_i are those of Section 2.1, so that both stationary curves live on the same grid. On each cell the singular factor is integrated with the closed form

$$\int_{\ell}^u |x-y|^p dy = \frac{1}{p+1} \begin{cases} (x-\ell)^{p+1} - (x-u)^{p+1}, & x \geq u, \\ (u-x)^{p+1} - (\ell-x)^{p+1}, & x \leq \ell, \\ (x-\ell)^{p+1} + (u-x)^{p+1}, & \ell < x < u, \end{cases} \quad p > -1. \quad (10)$$

On the exterior only the right half-line is discretized, since W is even; the left half-line enters through the mirrored kernels $G_\Omega^\sigma(y, -z)$ and $(y+x)^{-1-2\sigma}$. In the variable $t = z - L \in (0, Y_{\max} - L]$ the mesh consists of a first cell $[0, h/4]$ followed by geometrically graded cells with N_{dec} cells per decade up to $Y_{\max}$; the point $z = 1$, where $Q_{a,r}$ has its cusp, is inserted as a cell edge. With the default values $N_{\text{dec}} = 40$ and $Y_{\max} = 10^4$ this gives $N_e = 297$ exterior cells per side, with edges $z_{k-1} < z_k$ and midpoints $\bar{z}_k$.

Step 1: W at the exterior midpoints. Product integration is used: the smooth factor $\kappa_{1,\sigma} Q_{a,r}(z) B(\eta_I(\bar{z}_j, z)/(1 + \eta_I); \sigma, \frac{1}{2} - \sigma)$ is frozen at $z = \bar{z}_k$ on each cell and the singular factor is integrated exactly with (10),

$$W_j = \kappa_{1,\sigma} \sum_{k=1}^{N_e} Q_{a,r}(\bar{z}_k) \left[B_{jk}^+ \int_{z_{k-1}}^{z_k} |\bar{z}_j - z|^{r-1} dz + B_{jk}^- \int_{z_{k-1}}^{z_k} (\bar{z}_j + z)^{r-1} dz \right] + W_j^{\text{tail}}, \quad (11)$$

where $B_{jk}^\pm$ are the values (8) at $(\bar{z}_j, \pm \bar{z}_k)$. The diagonal cell $k = j$ is exact ($\eta_I = \infty$ and the third line of (10)). The tail $W_j^{\text{tail}} = \int_{Y_{\max}}^\infty [G(\bar{z}_j, z) + G(\bar{z}_j, -z)] Q_{a,r}(z) dz$ is computed with a 48-point Gauss–Legendre rule in the mapped variable $z = Y_{\max}(1 - s)^{-1/(1-a)}$, $s \in [0, 1]$, for which the integrand is bounded and smooth because $Q_{a,r}(z)|z|^{r-1} dz \sim z^{a-2} dz$.

Step 2: the density on I . Freezing W at the exterior midpoints and integrating the kernel exactly over each exterior cell,

$$\phi_i = Q_{a,r}(x_i) + C_{1,\sigma} \left[\sum_{j=1}^{N_e} W_j \left(\frac{(z_{j-1} - x_i)^{-r} - (z_j - x_i)^{-r}}{r} + \frac{(z_{j-1} + x_i)^{-r} - (z_j + x_i)^{-r}}{r} \right) + T_i \right], \quad (12)$$

where the tail $T_i = \int_{Y_{\max}}^\infty W(y) [(y - x_i)^{-1-r} + (y + x_i)^{-1-r}] dy$ uses the extrapolation $W(y) \approx W_{N_e}(y/\bar{z}_{N_e})^{a-1}$ with the exact decay exponent and a 48-point Gauss–Legendre rule in the variable $y = Y_{\max}(1 - s)^{-1/(1-\beta)}$. The kernel in (12) is never singular because $x_i \leq L - h/2 < L \leq z_{j-1}$; the first exterior cell of width $h/4$ resolves the near-singularity at the boundary cells. No clipping or normalization is applied to ϕ_i before the diagnostics are computed.

Diagnostics and output. The script prints the *raw mass* $M_G = h \sum_i \phi_i$ of the reconstruction, the mass of its negative part $M_- = h \sum_i \max(-\phi_i, 0)$, and the ratio between the computed tail coefficient $W_{N_e} \bar{z}_{N_e}^{1-a}$ and the asymptotic constant A of (9). Since (3) is an exact representation of the stationary probability and L has been selected by the mass condition, M_G must converge to one as the discretization is refined; this is a genuine test of the constants, because a wrong relative weight of source and correction would change M_G (the correction contributes ≈ -0.13 of the total mass) and would produce the wrong shape. With the default parameters $M_G = 0.99990$, $M_- = 0$, and the tail-coefficient ratio is 0.99967; the convergence of M_G to one under refinement is reported in Tables 4–6. For the plot the density is clipped at zero (inactive here) and its CDF F_G is normalized by its final value. The reported discrepancy $\max_{0 \leq j \leq n} |F_G(e_j) - F_I(e_j)|$ compares the normalized CDFs of (7.3) and (7.4) on the same cell edges; with the default parameters it equals 5.4×10^{-5} . The maximum relative density difference $\max |\phi_i - c_i| / \max c_i$ over the central 90% of the cells is 7×10^{-4} ; near the endpoints, where both densities behave like $(L - |x|)^{-\sigma}$, the two discretizations differ by up to about 1% in the last cells.

Meaning of the output. This route produces the second stationary density and CDF in Figure 1. Unlike the zero-flux solve, it involves no least-squares problem: the density is assembled from the exterior source through two explicit quadratures with fixed analytic constants. Its agreement with the zero-flux density of Section 2.1, including the unnormalized mass, checks the formula (7.3) and the constants of Table 1 rather than the shape alone.

Remark. *The Boggio/Green formulation for H and the Poisson formulation for the σ -harmonic extension U of $g = V''/(r(1+r))$ are equivalent through the exact change of unknown $H = r(1+r)U - V''$, and both give $\phi_I = \gamma_{1,r}^{-1}(-\Delta)^\sigma U$ on I with $\gamma_{1,r}^{-1} = -c_{1,r}r(1+r) = 1/(2\Gamma(r)\cos(\pi r/2))$. The equivalence holds for the continuum solutions in matched classes at infinity; it does not assert equality of finite-resolution approximations. The script implements the Green formulation because it is the form stated in Proposition 7.1 and because it fixes all constants.*

2.3 Particle dynamics and diagnostics

Original problem and purpose. The evolution equation transports a probability measure with velocity $-K_a * \omega_m + K_r * \phi_t$. Replacing that measure by N equally weighted point masses yields the particle ODE below. Starting away from the computed stationary profile and comparing its distribution at $T = 50$ with Sections 2.1–2.2 tests whether this trajectory approaches the predicted profile in the displayed experiment.

Numerical calculation. The particle system has $N = 2000$, time step $\Delta t = 0.02$, and 2500 explicit Euler steps, giving $T = 50$. With `rng(12)`, its initial positions are $X_i^0 = -1.4 + 2.8(i-1)/(N-1) + 0.01\xi_i$, where ξ_i are the values returned by `randn`; the initial positions are sorted. The exact background force is $F_\omega(x) = \frac{m}{2}(|x+1|^{1+a} - |x-1|^{1+a})$, and the update is

$$X_i^{k+1} = X_i^k + \Delta t \left(-F_\omega(X_i^k) + \frac{1+r}{N} \sum_j \text{sgn}(X_i^k - X_j^k) |X_i^k - X_j^k|^r \right).$$

All pairwise terms use the old positions. The diagonal is explicitly set to zero. No force smoothing is used. Positions are sorted every 20 steps and after the final step. Termination is at the fixed time T , with no residual-based early stopping.

Every four steps the script records the diameter $D = \max_i X_i - \min_i X_i$ and twice the 99% quantile of $|X_i|$; the final diameter is printed together with $2L$. For the empirical CDF $F_N(x, T) = N^{-1} \sum_i \mathbf{1}\{X_i(T) \leq x\}$ and the zero-flux CDF F_I of Section 2.1, the reported error is $E_{\text{CDF}} = \max_{0 \leq j \leq 320} |F_N(e_j, T) - F_I(e_j)|$, a maximum on the selected cell edges. The histogram uses 55 equally spaced edges (54 bins) on $[-1.05L, 1.05L]$ and `Normalization='pdf'`; with $N = 2000$ each bin inside the support contains about 37 particles, so the bin-to-bin fluctuation of the histogram is about 3% (it was 11% with $N = 500$).

Equilibration. At $T = 50$ the particle system has reached the equilibrium configuration of the N -particle ODE to machine precision: Table 7 shows that E_{CDF} and the final diameter are identical to eight digits for $\Delta t \in \{0.08, 0.04, 0.02, 0.01\}$ and for $T \in \{50, 100, 150, 200\}$. The remaining discrepancy against the zero-flux CDF is therefore the discretization error of the N -particle approximation, and it decreases like $1/N$. As a consequence the printed values do not depend on the initial perturbation and hence not on the random number generator.

2.4 Reading the one-dimensional figure

Left picture: densities and particle histogram. The horizontal axis is position x and the vertical axis is probability density. The solid curve is ϕ_I from the zero-flux condition (7.4) (Section 2.1), the dash-dotted curve ϕ_I from the Green-kernel reconstruction (7.3) (Section 2.2), and the dashed histogram is formed from the evolved particles at $T = 50$. The vertical dotted lines mark the selected endpoints $\pm L$ with $L \approx 0.2727$. The reference density rises near these endpoints (like $(L - |x|)^{-\sigma}$ with $\sigma = 0.1$) and is lower near the center; the two stationary curves are indistinguishable at the resolution of the figure, and the histogram follows this shape with bin-to-bin fluctuations expected for finitely many particles. This panel compares the spatial density profile, not the background density.

Middle picture: three cumulative distributions. At position x , each curve gives the probability mass to the left of x . The three curves are the integrated zero-flux density (7.4), the normalized Green-kernel reconstruction (7.3), and the empirical CDF of the particles at $T = 50$. Their overlap shows agreement of the two stationary-profile computations with the evolved particle distribution. The figure title reports the two maximum CDF discrepancies on the cell edges, 5.4×10^{-5} between the two stationary curves and 2.5×10^{-4} between the particles and the zero-flux curve; neither is an error against an exact solution.

Right picture: zero-flux residual versus the prescribed half-width. The horizontal axis is the prescribed half-width L of the zero-flux system of Section 2.1, the vertical (logarithmic) axis the RMS force residual of its nonnegative least-squares solution. The residual decreases steeply towards the selected $L \approx 0.2727$ and increases again beyond it: to the left the interval is too short for the repulsion to balance the attraction, to the right the least-squares solution empties the outermost cells and the force at the outer test points cannot be balanced. The square marker is the minimizer found by `fminbnd`. This panel shows that the support is determined by the zero-flux condition itself; it is the one-dimensional counterpart of the right panel of Figure 2. The particle diameter, which is not shown, is 0.54507 at $T = 50$, i.e. $2L - 3.0 \times 10^{-4}$: the outermost particles sit slightly inside the support of the continuum solution, as expected for finitely many particles.

3 Two-dimensional radial experiment

3.1 Solving the two-dimensional stationary radial force balance

Original problem. For the Gaussian background, seek a radial stationary probability measure supported in a disk of radius R . Write the radial mass measure as $d\mu(\eta) = 2\pi\eta\phi(\eta)d\eta$ when it has a density. The force condition (7.13) and the mass constraint read

$$\int_0^R \mathcal{A}_r(\rho, \eta) d\mu(\eta) = m \int_0^\infty \mathcal{A}_a(\rho, \eta) \eta e^{-\eta^2/2} d\eta, \quad 0 < \rho < R, \quad \mu([0, R]) = 1,$$

with

$$\mathcal{A}_\alpha(\rho, \eta) = \frac{1 + \alpha}{2\pi} \int_0^{2\pi} (\rho - \eta \cos \theta)(\rho^2 + \eta^2 - 2\rho\eta \cos \theta)^{(\alpha-1)/2} d\theta.$$

To derive these expressions, place the observation point at $x = (\rho, 0)$ and a source point at $y = (\eta \cos \theta, \eta \sin \theta)$; the radial component of $K_\alpha(x - y) = (1 + \alpha)(x - y)|x - y|^{\alpha-1}$ is $(1 + \alpha)(\rho - \eta \cos \theta)(\rho^2 + \eta^2 - 2\rho\eta \cos \theta)^{(\alpha-1)/2}$, and averaging over the angle gives $\mathcal{A}_\alpha$; tangential components cancel by symmetry. The kernel $\mathcal{A}_r$ acts on the unknown stationary radial mass, whereas $\mathcal{A}_a$ acts on the known Gaussian background radial mass $m\eta e^{-\eta^2/2} d\eta$.

Why solve for ring masses? Each unknown is a mass distributed uniformly around a circle. Its force contribution is its mass times $\mathcal{A}_r$, so the force equation is linear in the unknown ring masses. Nonnegative least squares imposes nonnegativity, while an appended weighted equation encourages unit total mass. Using more test radii than unknown rings makes the force system overdetermined and provides sampled residuals. The radius is prescribed in each solve and selected afterwards as the residual minimizer, exactly as L in Section 2.1.

Numerical calculation. Here $(a, r, m) = (0.6, 0.2, 1.8)$ and $\omega_m(x) = \frac{m}{2\pi} e^{-|x|^2/2}$. The radial force of a unit circular mass is approximated as

$$\mathcal{A}_{\alpha,q}(\rho, s) = \frac{1 + \alpha}{q} \sum_{\ell=1}^q (\rho - s \cos \theta_\ell) (\max(\rho^2 + s^2 - 2\rho s \cos \theta_\ell, 10^{-14}))^{(\alpha-1)/2}, \quad \theta_\ell = \frac{2\pi(\ell - \frac{1}{2})}{q}.$$

The base discretization uses $n_{\text{ring}} = 110$ source rings at $s_j = (j - \frac{1}{2})R/110$, $n_{\text{test}} = 150$ test radii $\rho_i = (i - \frac{1}{2})R/150$, and $q = 160$ angular nodes. The background uses $n_{\text{bg}} = 550$ radial midpoint nodes on $[0, 6]$ with weights $w_\ell^b = m s_\ell^b e^{-(s_\ell^b)^2/2} \Delta s^b$, not renormalized. Set $A_{ij} = \mathcal{A}_{r,160}(\rho_i, s_j)$ and $b_i = \sum_\ell \mathcal{A}_{a,160}(\rho_i, s_\ell^b) w_\ell^b$. The code solves $\min_{\mu \geq 0} \{\|A\mu - b\|_2^2 + 120^2(\mathbf{1}^\top \mu - 1)^2\}$ with `lsqnonneg` (default settings); if `lsqnonneg` is unavailable, 800 projected-gradient iterations on the augmented system are used instead. The result is clipped and divided by its total mass; the final solution defines $\text{Res} = A\mu - b$, and the console reports $\|\text{Res}\|_2/\sqrt{150}$ and $\|\text{Res}\|_\infty$.

Selection of the radius. The RMS residual is scanned over $R \in \{0.260, 0.2625, \dots, 0.400\}$ and minimized with `fminbnd` on the bracket around the smallest scan value (tolerance 10^{-6}), giving

$$R = 0.32845, \quad \|\text{Res}\|_2/\sqrt{150} = 6.0 \times 10^{-7}, \quad \|\text{Res}\|_\infty = 4.4 \times 10^{-6}.$$

The residual is V -shaped in R (right panel of Figure 2 and Table 2): at $R = 0.33$ it is 5.4×10^{-6} , at $R = 0.32$ and 0.34 it is larger by two orders of magnitude, and for $R \geq 0.36$ the nonnegative solution empties the outer rings altogether (last column of Table 2), because the support of the balanced measure is smaller than the prescribed disk and the outer test radii then lie outside the actual support. The figures and all subsequent two-dimensional calculations use the minimizer $R \approx 0.33$.

Character of the solution. The stationary radial density is strongly singular at the boundary (its cumulative mass rises steeply near R ; at the base level 0.517 of the mass lies in the annulus $0.9R < \rho \leq R$). The nonnegative least-squares ring masses resolve this boundary layer in the typical oscillatory way of NNLS solutions of first-kind equations: neighbouring outer rings carry alternately large and small masses, and a few rings carry none. The sampled force residual is nevertheless small, because the force is a smoothing functional of the mass. This has a consequence for comparisons: pointwise (sup-norm) differences between cumulative radial masses are dominated by where exactly the staircase jumps inside the last two per cent of the radius and do not decrease under refinement, whereas the L^1 distance $W_1(F_1, F_2) = \int_0^{0.66} |F_1(\rho) - F_2(\rho)| d\rho$ (the Wasserstein-1 distance between the radial mass distributions) measures the mass allocation itself. The script prints both; Section 4 uses W_1 for the two-dimensional comparisons.

Meaning of the output. The normalized ring masses define the first stationary radial CDF in Figure 2 and the selected radius; the residual measures the remaining force imbalance at the test radii after normalization. This calculation solves the force-verification equation (7.13) directly and does not evaluate the exterior Green formula.

3.2 The disk formula (7.11)–(7.12) with analytic constants

Original problem. The second stationary curve in Figure 2 is the Green-kernel reconstruction of Proposition 7.3. With $s = (1+r)/2 = 0.6$, $Q_{a,r} = \mathcal{L}_r V$ the source (6.10) and $c_{2,r} = \gamma_{2,r}^{-1}$,

$$\phi_R = (Q_{a,r} - c_{2,r}(-\Delta)^s H_R)|_{B_R}, \quad H_R(y) = c_{2,r}^{-1} \int_{B_R^c} G_R^s(y, z) Q_{a,r}(z) dz \quad (y \in B_R^c), \quad H_R = 0 \text{ in } B_R, \quad (13)$$

where by Boggio's formula (6.20) for the exterior of the disk

$$G_R^s(y, z) = \kappa_{2,s} |y-z|^{2s-2} \int_0^{\eta_R(y,z)} \frac{t^{s-1}}{1+t} dt, \quad \eta_R(y, z) = \frac{(|y|^2 - R^2)(|z|^2 - R^2)}{R^2 |y-z|^2}, \quad \kappa_{2,s} = \frac{1}{4^s \pi \Gamma(s)^2}. \quad (14)$$

The implementation evaluates (13) exactly as the one-dimensional routine of Section 2.2 evaluates (7.3): all constants are those of Table 1, nothing is fitted, and the unnormalized mass of ϕ_R is a diagnostic.

The same three reductions as in one dimension. (i) Since $H_R = 0$ in B_R , for $x \in B_R$

$$(-\Delta)^s H_R(x) = -C_{2,s} \int_{B_R^c} \frac{H_R(y)}{|x-y|^{2+2s}} dy, \quad C_{2,s} = \frac{4^s \Gamma(1+s)}{\pi |\Gamma(-s)|}, \quad (15)$$

with no principal value ($|x-y| \geq R-|x| > 0$ on B_R^c), so that with $W := c_{2,r} H_R = \int_{B_R^c} G_R^s(\cdot, z) Q_{a,r}(z) dz$ the constant $c_{2,r}$ cancels and

$$\phi_R(x) = Q_{a,r}(x) + C_{2,s} \int_{B_R^c} \frac{W(y)}{|x-y|^{2+2s}} dy, \quad x \in B_R. \quad (16)$$

Numerically $\kappa_{2,s} = 6.2530 \times 10^{-2}$, $C_{2,s} = 0.17670$ and $B(s, 1-s) = \pi / \sin(\pi s) = 3.3029$.

(ii) *The source in closed form.* For the Gaussian background $\omega_m = \frac{m}{2\pi} e^{-|x|^2/2}$ one has $\widehat{\omega_m}(\xi) = m e^{-|\xi|^2/2}$, and the Fourier form (6.9) of the source, $\widehat{Q_{a,r}} = (\gamma_{2,a}/\gamma_{2,r}) |\xi|^{-\beta} \widehat{\omega_m}$ with $\beta = a - r = 0.4$, gives by radial inversion the Hankel integral

$$Q_{a,r}(\rho) = \frac{\gamma_{2,a}}{\gamma_{2,r}} \frac{m}{2\pi} \int_0^\infty k^{1-\beta} e^{-k^2/2} J_0(k\rho) dk = \frac{\gamma_{2,a}}{\gamma_{2,r}} \frac{m}{2\pi} 2^{-\beta/2} \Gamma(\alpha) {}_1F_1\left(\alpha; 1; -\frac{\rho^2}{2}\right), \quad \alpha = 1 - \frac{\beta}{2} = 0.8, \quad (17)$$

with the ratio $\gamma_{2,a}/\gamma_{2,r} = 2^\beta \Gamma(\frac{3+a}{2}) \Gamma(-\frac{1+r}{2}) / (\Gamma(-\frac{1+a}{2}) \Gamma(\frac{3+r}{2}))$ from Table 1. The Kummer function is evaluated from its power series in the form ${}_1F_1(\alpha; 1; -z) = e^{-z} {}_1F_1(1-\alpha; 1; z)$ (all terms positive) for $z = \rho^2/2 < 40$, and from its large- z expansion ${}_1F_1(\alpha; 1; -z) \sim z^{-\alpha} \Gamma(1-\alpha)^{-1} \sum_k (\alpha)_k^2 / (k! z^k)$ otherwise. The script checks (17) against direct quadrature of the Hankel integral (agreement to 10^{-12}) and against the Riesz-potential form (6.10): as $\rho \rightarrow \infty$, $Q_{a,r}(\rho) \rho^{2-\beta} \rightarrow C_{2,a,r} m$ with $C_{2,a,r}$ from Table 1 (agreement to 10^{-6} at $\rho = 10^3$), which confirms both the constant and the normalization of the Hankel form. $Q_{a,r}$ is positive, radially decreasing, $Q_{a,r}(0) = 0.2573$, and decays like $\rho^{-(2-\beta)} = \rho^{-1.6}$.

(iii) *Boggio's integral* is again an incomplete beta function, $\int_0^\eta t^{s-1} (1+t)^{-1} dt = B(s, 1-s)$ with the argument formed as $N/(N+R^2|y-z|^2)$, $N = (|y|^2 - R^2)(|z|^2 - R^2)$.

Radial reduction. All functions are radial, so both integrals in (16) reduce to one radial and one angular integration. With $d^2(\rho, \eta, \theta) = (\rho-\eta)^2 + 4\rho\eta \sin^2(\theta/2)$ (this form avoids cancellation near the diagonal),

$$W(\rho) = \int_R^\infty Q_{a,r}(\eta) \mathcal{K}_G(\rho, \eta) \eta d\eta, \quad \mathcal{K}_G(\rho, \eta) = \kappa_{2,s} B(s, 1-s) \int_0^{2\pi} d^{2s-2} I_{N/(N+R^2 d^2)}(s, 1-s) d\theta, \quad (18)$$

$$\phi_R(\rho) = Q_{a,r}(\rho) + C_{2,s} \int_R^\infty W(\eta) \mathcal{K}_2(\rho, \eta) \eta d\eta, \quad \mathcal{K}_2(\rho, \eta) = \int_0^{2\pi} d^{-2-2s} d\theta. \quad (19)$$

The angular integrands are peaked at $\theta = 0$: $d^{2s-2} = d^{-0.8}$ is integrable but singular on the diagonal $\rho = \eta$, and $d^{-2-2s} = d^{-3.2}$ is nearly singular when ρ and η are both close to R . Both are integrated with a Gauss–Legendre rule in the graded variable $\theta = \pi u^5$, $u \in (0, 1)$, $n_{\text{ang}} = 64$ nodes, doubled for the two half circles; on the diagonal the integrand times $d\theta/du$ is bounded, and for a peak of relative width $\delta = |\rho-\eta|/\sqrt{\rho\eta}$ the peak occupies a u -interval of width $(\delta/\pi)^{1/5}$, which 64 nodes resolve down to $\delta \sim 10^{-6}$.

Behaviour at the boundary and at infinity. As in one dimension, $W \sim A\eta^{a-1} = A\eta^{-0.4}$ at infinity (from $Q \sim \eta^{\beta-2}$ and $G \sim \eta^{2s-2}$), so the η -integral of (18) converges only like $Y_{\text{max}}^{-0.4}$ (without tail correction the mass is 0.93 at $Y_{\text{max}} = 10^2$ and 0.99 at 10^4), whereas the η -integral of (19) converges like $Y_{\text{max}}^{-1.6}$. Near the boundary $H_R \sim \text{dist}(\cdot, B_R)^s$ from outside, hence by (15) the density behaves like

$$\phi_R(\rho) \sim c(R-\rho)^{-s}, \quad \rho \rightarrow R^-, \quad (20)$$

an integrable but strong singularity ($s = 0.6$), which is the two-dimensional analogue of $(L - |x|)^{-\sigma}$ and explains why about half of the mass sits in the outer ten per cent of the radius. The script fits the exponent of the computed density on $10^{-5} < (R - \rho)/R < 10^{-2}$ and obtains -0.599 , confirming (20); the exponent is used only to integrate the density exactly across the cells (below), not to modify the density.

Numerical calculation. The radius is the minimizer R of Section 3.1. *Exterior mesh:* in $t = \eta - R$, a first cell $[0, 10^{-7}R]$ followed by geometrically graded cells with $N_{\text{dec}} = 40$ cells per decade up to $Y_{\text{max}} = 10^4$ ($N_e = 460$ cells with edges $\eta_{k-1} < \eta_k$ and midpoints $\bar{\eta}_k$); the fine resolution at the boundary is needed because the kernel $\mathcal{K}_2(\rho, \eta)$ varies on the scale $\eta - R$ when ρ is close to R . *Interior mesh:* cells whose distance to the boundary $R - \rho$ spans seven decades with $I_{\text{dec}} = 40$ cells per decade, plus one cell at the centre ($n_I = 281$ cells with midpoints $\bar{\rho}_i$). *Step 1:* W at the exterior midpoints by the midpoint rule in η with the angular integrals (18) evaluated at every pair $(\bar{\eta}_j, \bar{\eta}_k)$,

$$W_j = \sum_{k=1}^{N_e} Q_{a,r}(\bar{\eta}_k) \mathcal{K}_G(\bar{\eta}_j, \bar{\eta}_k) \bar{\eta}_k (\eta_k - \eta_{k-1}) + W_j^{\text{tail}},$$

where the diagonal term is finite because the angular integral of $d^{-0.8}$ is finite, and the tail over (Y_{max}, ∞) is a 48-point Gauss–Legendre rule in $z = Y_{\text{max}}(1 - \varsigma)^{-1/(1-a)}$. *Step 2:* the density at the interior midpoints by (19) with the same midpoint rule and with the tail computed from the extrapolation $W(\eta) \approx W_{N_e}(\eta/\bar{\eta}_{N_e})^{a-1}$ and a mapped 48-point rule. *Step 3:* ring masses. Because of (20), the midpoint rule would integrate the density poorly over the outer cells; the mass of cell i is therefore computed by product integration with the boundary weight,

$$\nu_i = \phi_R(\bar{\rho}_i)(R - \bar{\rho}_i)^s \int_{\rho_{i-1}}^{\rho_i} (R - \rho)^{-s} 2\pi\rho \, d\rho,$$

the integral being elementary. The raw mass is $M_G = \sum_i \nu_i$; the plotted CDF is the cumulative sum of the normalized ν_i . Since the same closed forms and quadratures are used for every cell, no coefficient, floor or cut-off enters except Y_{max} , whose influence is removed by the tail terms.

Diagnostics and output. At the selected R and the default resolution the script prints $M_G = 1.0028$ and $M_- = 0$; M_G moves towards one under refinement of the exterior mesh (1.0059, 1.0028, 1.0012 for 20, 40, 80 cells per decade, Table 9), the remaining deviation being the discretization error of the radial midpoint rule at the cusp of $\mathcal{K}_G$ on the diagonal. The normalized ring masses are also inserted into the force equation (7.13) of Section 3.1; their RMS residual, 3.7×10^{-4} , is an independent measure of how well the reconstructed density balances the forces (the ring-mass solution has a residual of 6×10^{-7} because it minimizes exactly this quantity). The W_1 distance between the reconstructed CDF and the ring-mass CDF is 1.7×10^{-3} , comparable to the W_1 self-difference of the ring solution between refinement levels (Table 8).

Meaning of the output. This route produces the second stationary radial CDF in Figure 2 from the exterior source alone, through two explicit quadratures with the analytic constants of Table 1, and independently of the ring-mass least-squares solve. Its agreement with the ring masses in mass, in W_1 and in the force residual checks the formula (7.11)–(7.12) and the constants; the boundary exponent (20) is a further qualitative confirmation. What is not resolved to high accuracy in either route is the individual mass allocation inside the boundary layer, cf. Section 3.1.

3.3 Radial evolution and visualization

Original problem and purpose. For a radial solution, the characteristic radius satisfies the background attraction plus the self-repulsion averaged over angles. Approximating the evolving radial mass by equal-weight circular masses yields the ODE discretized below. Its final radial CDF is compared with the stationary CDFs from Sections 3.1–3.2. The Cartesian cloud is a visualization of those final radii.

Numerical calculation. The dynamics uses $N = 180$ equal-weight radial nodes, initialized deterministically by $\rho_i^0 = 1.1\sqrt{-2\log(1 - 0.98u_i)}$, $u_i = (i - \frac{1}{2})/N$. The force tables use 260 uniform radii on $[0, 3.2]$ and 96 angular nodes; the background table uses the same 550 background nodes. Background queries use linear interpolation with linear extrapolation; self-force queries use bilinear interpolation with zero outside the table domain. The explicit Euler update is

$$\rho_i^{k+1} = \max\left\{\rho_i^k + \Delta t\left(-\tilde{F}_\omega(\rho_i^k) + \frac{1}{N} \sum_j \tilde{\mathcal{A}}_r(\rho_i^k, \rho_j^k)\right), 0\right\}.$$

The diagonal ring contribution is retained (a radial node represents mass uniformly distributed over a circle). Radii are sorted after every step. The time step is 0.05 and 1000 steps give $T = 50$. Stationary cumulative masses are evaluated on 500 radii in $[0, 0.66]$ and compared with the empirical radial CDF; the script prints $\sup_\rho |\Delta F|$ and W_1 against the ring-mass curve. The cloud visualization assigns angles $\vartheta_i = (i - 1)\pi(3 - \sqrt{5}) \bmod 2\pi$; these angles are for display only.

3.4 Reading the two-dimensional figure

Left picture. Each marker is placed at an evolved radius and a deterministic display angle. The dashed circle marks the selected support boundary $R \approx 0.33$. The points occupy the disk and reach close to this circle; the picture is a qualitative support check.

Middle picture. The horizontal axis is radius ρ and the vertical axis the total probability inside B_ρ . The solid curve is obtained from the zero-flux ring-mass solve (7.13) of Section 3.1, the dash-dotted curve from the disk formula (7.11) of Section 3.2, and the dashed staircase is the empirical radial mass at $T = 50$. The vertical dotted line marks $R \approx 0.33$. The curves overlap, rise steeply near the support boundary, and attain mass one near R ; this indicates agreement in how mass is distributed by radius, including the large mass near the outer part of the disk. The two stationary curves are computed independently (least squares on the force equation versus explicit quadrature of the exterior formula with fixed constants); the pointwise difference inside the boundary layer is not small because the ring-mass staircase is oscillatory there (Section 3.1), so the agreement visible in the figure is an agreement of mass allocation, quantified by W_1 .

Right picture: zero-flux residual versus the prescribed radius. The horizontal axis is the prescribed disk radius R of the ring-mass system of Section 3.1, the vertical (logarithmic) axis the RMS force residual of its nonnegative least-squares solution, and the square marker the minimizer. As in one dimension the residual is V-shaped: it decreases by four orders of magnitude towards $R \approx 0.33$ and increases again beyond it. Note that this is a curve over the *support* radius, one solve per point; the residual of one fixed solve as a function of the *test* radius (printed by the script as RMS and maximum values) is small over most of the disk and increases only towards the boundary layer, and would not display a minimum. The curve belongs to the zero-flux ring solution, not to the particle evolution and not to the Green curve.

4 Refinement study

Part III of the script repeats each computation of Sections 2–3 for a sequence of discretization parameters and terminal times and records how the outputs change. Two kinds of quantities are reported: *self-convergence* of one method (differences between successive resolutions or against the finest resolution), and *cross-method discrepancies* (Green versus zero-flux, particles versus stationary curves). All tables below are written by the script into `refinement_tables.tex` and included here unchanged; the corresponding log is `refinement_results.txt`. Figure 1 collects the convergence plots.

4.1 Selection of the support parameters

Tables 1 and 2 list the RMS force residual of the zero-flux systems as functions of the prescribed half-width L (one dimension, $n = 320, 240$ test points) and of the prescribed radius R (two dimensions, base level). Both curves are V -shaped with a minimum four orders of magnitude below the values at the ends of the scanned range, which is what selects $L \approx 0.2727$ and $R \approx 0.33$; the last line of each table gives the minimizer found by `fminbnd`. In one dimension the minimizer differs by 2.9×10^{-5} from the half-width obtained from the mass condition of the differentiated form (7.2). In two dimensions, the mass of the solution in the outer annulus $(0.9R, R]$ shows how the nonnegative solution reacts to a wrong radius: for R too small more than half of the mass is pushed against the boundary, for $R \geq 0.36$ the outer rings are emptied.

L	RMS residual	L	RMS residual	L	RMS residual
0.2400	2.32×10^{-3}	0.2650	5.06×10^{-4}	0.2900	4.73×10^{-3}
0.2425	2.13×10^{-3}	0.2675	3.35×10^{-4}	0.2925	5.84×10^{-3}
0.2450	1.94×10^{-3}	0.2700	1.71×10^{-4}	0.2950	7.02×10^{-3}
0.2475	1.76×10^{-3}	0.2725	1.18×10^{-5}	0.2975	8.28×10^{-3}
0.2500	1.57×10^{-3}	0.2750	2.00×10^{-4}	0.3000	9.61×10^{-3}
0.2525	1.39×10^{-3}	0.2775	6.33×10^{-4}	0.3025	1.10×10^{-2}
0.2550	1.21×10^{-3}	0.2800	1.22×10^{-3}	0.3050	1.25×10^{-2}
0.2575	1.03×10^{-3}	0.2825	1.94×10^{-3}	0.3075	1.40×10^{-2}
0.2600	8.54×10^{-4}	0.2850	2.77×10^{-3}	0.3100	1.55×10^{-2}
0.2625	6.78×10^{-4}	0.2875	3.71×10^{-3}		
$L^* = 0.272685$	2.45×10^{-7}	L from the mass condition of (7.2): 0.272657			

Table 1: One dimension: RMS force residual of the zero-flux system (1) as a function of the prescribed half-width L ($n = 320$ cells, 240 test points), and the minimizer.

4.2 One-dimensional zero-flux state (7.4)

For $n \in \{40, 80, \dots, 1280\}$ (with $3n/4$ test points) the half-width L_n is recomputed as the residual minimizer on the n -cell grid, and the density ϕ_n and its CDF F_n are compared with the $n = 1280$ solution (F_n extended by 0 and 1 outside $[-L_n, L_n]$ and compared on a uniform grid of 6001 points in $[-0.3, 0.3]$). Table 3 shows first-order convergence of L_n and of the CDF, consistent with the piecewise-constant approximation of a density with the endpoint singularity $(L - |x|)^{-\sigma}$; the central density value converges about quadratically, and the RMS residual decreases like $n^{-1.6}$. The half-width from the mass condition of the differentiated form (7.2), listed in the third column, approaches the residual minimizer as n grows (the difference decreases from 2.3×10^{-4} at $n = 40$ to 7×10^{-6} at $n = 1280$), which confirms that the two formulations discretize the same stationary state.

R	RMS residual	mass in $(0.9R, R]$	R	RMS residual	mass in $(0.9R, R]$
0.260	8.60×10^{-3}	0.9671	0.335	7.39×10^{-5}	0.4657
0.265	7.13×10^{-3}	0.9514	0.340	2.50×10^{-4}	0.4576
0.270	5.86×10^{-3}	0.9331	0.345	5.35×10^{-4}	0.4227
0.275	4.76×10^{-3}	0.9122	0.350	9.29×10^{-4}	0.4524
0.280	3.82×10^{-3}	0.8883	0.355	1.43×10^{-3}	0.1164
0.285	3.01×10^{-3}	0.8613	0.360	2.04×10^{-3}	0.0000
0.290	2.32×10^{-3}	0.8307	0.365	2.74×10^{-3}	0.0000
0.295	1.74×10^{-3}	0.7959	0.370	3.55×10^{-3}	0.0000
0.300	1.26×10^{-3}	0.7562	0.375	4.46×10^{-3}	0.0000
0.305	8.71×10^{-4}	0.7103	0.380	5.47×10^{-3}	0.0000
0.310	5.62×10^{-4}	0.6566	0.385	6.57×10^{-3}	0.0000
0.315	3.25×10^{-4}	0.5919	0.390	7.77×10^{-3}	0.0000
0.320	1.54×10^{-4}	0.5131	0.395	9.06×10^{-3}	0.0000
0.325	4.23×10^{-5}	0.5366	0.400	1.04×10^{-2}	0.0000
0.330	5.39×10^{-6}	0.4943			
$R^* = 0.32845$	6.03×10^{-7}				

Table 2: Two dimensions: RMS force residual of the ring-mass system (7.13) as a function of the prescribed radius R (every second scan point; $n_{\text{ring}} = 110$, $n_{\text{test}} = 150$, $q = 160$), the mass in $(0.9R, R]$, and the minimizer.

n	L_n	L_n from (7.2)	RMS residual	$\phi_n(0)$	$ L_n - L_{1280} $	$\max_x F_n - F_{1280} $
40	0.2732958	0.2730682	6.82×10^{-6}	1.723672	6.76×10^{-4}	1.53×10^{-3}
80	0.2729477	0.2728333	2.25×10^{-6}	1.723579	3.28×10^{-4}	7.89×10^{-4}
160	0.2727729	0.2727155	7.43×10^{-7}	1.723556	1.53×10^{-4}	3.91×10^{-4}
320	0.2726852	0.2726565	2.45×10^{-7}	1.723550	6.58×10^{-5}	1.77×10^{-4}
640	0.2726414	0.2726270	8.08×10^{-8}	1.723548	2.19×10^{-5}	6.19×10^{-5}
1280	0.2726194	0.2726122	2.67×10^{-8}	1.723548	—	—

Table 3: Zero-flux state (7.4): dependence on the number of cells n . L_n is the residual minimizer on each grid, the third column the half-width selected by the mass condition of (7.2) on the same grid, F_n the normalized CDF.

4.3 One-dimensional Green-kernel reconstruction (7.3)

The reconstruction depends on the exterior resolution N_{dec} (cells per decade), the truncation Y_{max} , and the interior cells n . In all cases L is the half-width of Figure 1 and the comparison partner is the zero-flux density (7.4) on the same n cells, so that only discretization differences are measured; M_G is the raw mass of the reconstruction and M_- the mass of its negative part.

Exterior resolution (Table 4). Already 10 cells per decade give M_G within 10^{-3} of one; from 40 cells per decade on, M_G and both discrepancies are converged to about 10^{-5} , and the remaining differences are those of the interior grid.

Truncation (Table 5). With the tail terms of Section 2.2, $Y_{\text{max}} = 10^2$ already gives M_G within 2×10^{-4} and $Y_{\text{max}} \geq 10^3$ is converged. Without the tail terms the raw mass is far from one and converges to it only like $Y_{\text{max}}^{-0.2}$ (0.75 at $Y_{\text{max}} = 10$, 0.98 at $Y_{\text{max}} = 10^6$); the normalized CDF discrepancy is less sensitive because the omitted contribution is close to a constant shift of the density, but the mass diagnostic is lost. This documents why the slow tails must be treated analytically.

Interior cells (Table 6). Here each grid uses its own half-width L_n , the residual minimizer of Section 4.2, so that the zero-flux state and the reconstruction are consistent on every grid. Refining n drives M_G to one and reduces the discrepancy against the zero-flux density on the same grid, in the CDF as well as in the central density.

cells/decade	N_e	M_G	M_-	$\max F_G - F_I $	$\max \phi_G - \phi_I / \max \phi_I$
5	39	0.99676259	0	5.88×10^{-5}	2.79×10^{-3}
10	76	0.99920438	0	5.54×10^{-5}	1.15×10^{-3}
20	149	0.99977752	0	5.43×10^{-5}	7.82×10^{-4}
40	297	0.99989990	0	5.39×10^{-5}	7.09×10^{-4}
80	592	0.99991837	0	5.38×10^{-5}	7.01×10^{-4}
160	1181	0.99991318	0	5.37×10^{-5}	7.08×10^{-4}

Table 4: Green-kernel reconstruction (7.3): exterior cells per decade at $n = 320$, $Y_{\max} = 10^4$. F_G , F_I are the normalized CDFs of the reconstruction and of the zero-flux state; the density discrepancy is taken over the central 90% of the cells.

$Y_{\max}$	N_e	M_G	$\max F_G - F_I $	M_G (no tail)	$\max F_G - F_I $ (no tail)
10^1	176	0.99824846	5.68×10^{-5}	0.754360	6.31×10^{-5}
10^2	217	0.99974825	5.42×10^{-5}	0.846483	5.61×10^{-5}
10^3	257	0.99989155	5.39×10^{-5}	0.903819	5.38×10^{-5}
10^4	297	0.99989990	5.39×10^{-5}	0.939576	5.33×10^{-5}
10^5	337	0.99989716	5.39×10^{-5}	0.961956	5.33×10^{-5}
10^6	377	0.99989471	5.39×10^{-5}	0.976004	5.35×10^{-5}

Table 5: Green-kernel reconstruction (7.3): exterior truncation at $n = 320$, 40 cells per decade, with and without the two tail terms.

4.4 One-dimensional particles

Table 7 varies the number of particles ($N \leq 2000$) at $\Delta t = 0.02$, $T = 50$; the time step at $N = 500$, $T = 50$; and the terminal time $T \in \{50, 100, 150, 200\}$ at $N = 500$, $\Delta t = 0.02$. The error E_{CDF} against the zero-flux CDF ($n = 320$) halves when N doubles, and the final diameter approaches $2L$ from below like $1/N$. Neither the time step nor the terminal time has any influence: all four time steps and all four terminal times give identical digits, because by $T = 50$ the explicit Euler iteration has converged to the equilibrium of the N -particle system, so that the terminal state is a fixed point independent of Δt and stationary in T . The particle curve in Figure 1 is thus a converged 2000-particle equilibrium, and its distance from the continuum profile is the $O(1/N)$ particle-discretization error.

4.5 Two-dimensional ring refinement

Table 8 doubles all four discretization parameters of the ring-mass system from level to level at the selected R . The RMS residual stays at a few 10^{-7} and the maximum residual, located at the outermost test radius, at a few 10^{-6} (the radius was selected at level 2, so the residual does not decrease monotonically at the finer levels); the W_1 distance between successive levels decreases, whereas the sup-norm distance stays of order 0.1 for the reason explained in Section 3.1: the position of the individual staircase jumps inside the boundary layer is not a convergent quantity of the NNLS ring solution, but the mass allocation is. A sharper resolution of the boundary layer would require a discretization adapted to the singular density; this is outside the scope of the present verification.

Disk formula (Table 9). The reconstruction of Section 3.2 is refined in each of its four parameters separately. The raw mass converges towards one under refinement of the exterior radial mesh, is insensitive to the truncation once the tail terms are included (the mass without tails is far from one at small $Y_{\max}$), and is converged in the angular and interior resolution; the boundary exponent is -0.60 throughout; the W_1 distance to the ring-mass curve decreases with the interior resolution and saturates at the level of the ring solution's own uncertainty;

n	L_n	N_e	M_G	$\max F_G - F_I $	$\max \phi_G - \phi_I / \max \phi_I$
40	0.2732958	261	0.99971646	3.45×10^{-4}	4.56×10^{-3}
80	0.2729477	273	0.99978110	1.87×10^{-4}	2.60×10^{-3}
160	0.2727729	285	0.99984763	1.01×10^{-4}	1.39×10^{-3}
320	0.2726852	297	0.99989993	5.40×10^{-5}	7.09×10^{-4}
640	0.2726414	309	0.99993653	2.88×10^{-5}	3.44×10^{-4}
1280	0.2726194	321	0.99996065	1.54×10^{-5}	1.61×10^{-4}

Table 6: Green-kernel reconstruction (7.3) versus zero-flux state (7.4) on the same n -cell grid with the half-width L_n of Table 3 (40 cells per decade, $Y_{\max} = 10^4$).

N	E_{CDF}	$D(T) - 2L$	Δt	E_{CDF}	$D(T) - 2L$	T	E_{CDF}	$D(T) - 2L$
125	4.00×10^{-3}	-2.80×10^{-3}	0.08	1.00×10^{-3}	-7.38×10^{-4}	50	1.00×10^{-3}	-7.38×10^{-4}
250	1.99×10^{-3}	-1.39×10^{-3}	0.04	1.00×10^{-3}	-7.38×10^{-4}	100	1.00×10^{-3}	-7.38×10^{-4}
500	1.00×10^{-3}	-7.38×10^{-4}	0.02	1.00×10^{-3}	-7.38×10^{-4}	150	1.00×10^{-3}	-7.38×10^{-4}
1000	4.99×10^{-4}	-4.36×10^{-4}	0.01	1.00×10^{-3}	-7.38×10^{-4}	200	1.00×10^{-3}	-7.38×10^{-4}
2000	2.50×10^{-4}	-2.96×10^{-4}						

Table 7: One-dimensional particle system: dependence on N ($\Delta t = 0.02$, $T = 50$; left), on Δt ($N = 500$, $T = 50$; middle) and on T ($N = 500$, $\Delta t = 0.02$; right). $D(T)$ is the final diameter, $L = 0.2726852$.

the force residual of (7.13) decreases with the exterior resolution.

4.6 Two-dimensional radial particles

Table 10 varies the number of radial nodes ($\Delta t = 0.05$, $T = 50$), the time step ($N = 180$, $T = 50$) and the terminal time $T \in \{50, 100, 150, 200\}$ ($N = 180$, $\Delta t = 0.05$), and compares the empirical radial CDF with the base-level ring-mass curve in the W_1 distance. The time step has no visible influence. In contrast to one dimension, the radial system is not fully at rest at $T = 50$: between $T = 50$ and $T = 200$ the largest radius still drifts from 0.32391 to 0.32351 and the W_1 distance to the ring-mass curve changes from 2.6×10^{-3} to 3.1×10^{-3} , a slow rearrangement inside the boundary layer. The W_1 error decreases with N but saturates at about 2.4×10^{-3} ; this floor, like the drift in T , is of the same size as the W_1 distance between the base-level ring solution and its finest refinement (Table 8, dotted line in Figure 1), so the comparison is limited by the stationary reference and by the different way in which the two discretizations represent the boundary layer. The sup-norm error is a boundary-layer quantity as in Section 3.1 and is not reported.

4.7 Summary

In both dimensions the support is selected by the zero-flux condition itself, through the minimum of the force residual over the prescribed support parameter, and the resulting stationary measures are compared with the Green-kernel formulas of Propositions 7.1 and 7.3 and with long-time particle solutions. The one-dimensional verification is constant-free and quantitatively converged: the Green-kernel reconstruction (7.3) evaluated with the analytic constants of Table 1 reproduces the zero-flux density on the same grid to 5×10^{-5} in the CDF, with unnormalized mass 0.9999 converging to one under refinement, and the particle system reaches an N -particle equilibrium whose distance from the continuum profile is $O(1/N)$. In two dimensions the disk formula (7.11)–(7.12) is evaluated in the same constant-free way, with the radial reduction of Section 3.2: its unnormalized mass is within 0.3% of one at the default resolution and converges under exterior refinement, its boundary exponent is the predicted $-s$, and the

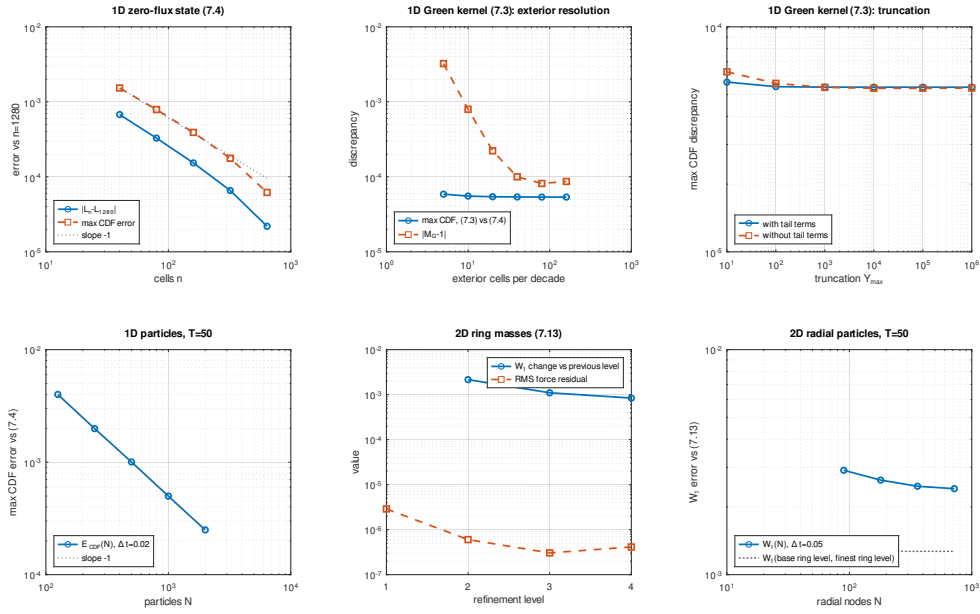


Figure 1: Refinement study produced by Part III of `matlabcode.m`. Top row: zero-flux state (7.4) (errors against $n = 1280$), Green-kernel reconstruction under exterior refinement, and under truncation with and without tail terms. Bottom row: one-dimensional particles versus N , W_1 change of the ring-mass CDF between levels together with the RMS residual, and two-dimensional radial particles versus N . Dotted lines indicate slope -1 or the reference level described in the text.

level	n_{ring}	n_{test}	q	n_{bg}	RMS residual	max residual	$\sup_{\rho} F_{\ell} - F_{\ell-1} $	$W_1(F_{\ell}, F_{\ell-1})$
1	55	75	80	275	2.89×10^{-6}	1.86×10^{-5}	–	–
2	110	150	160	550	6.03×10^{-7}	4.39×10^{-6}	2.65×10^{-1}	2.16×10^{-3}
3	220	300	320	1100	3.05×10^{-7}	3.27×10^{-6}	5.56×10^{-2}	1.10×10^{-3}
4	440	600	640	2200	4.13×10^{-7}	4.56×10^{-6}	1.66×10^{-1}	8.41×10^{-4}

Table 8: Zero-flux ring masses (7.13) at the selected R under simultaneous refinement of the rings, test radii, angular nodes and background nodes; F_{ℓ} is the cumulative radial mass at level ℓ .

parameter	value	N_e	n_I	M_G	M_G (no tail)	exponent	W_1	RMS residual
exterior cells/decade	10	116	281	1.01031	0.99841	-0.599	1.84×10^{-3}	1.05×10^{-3}
exterior cells/decade	20	231	281	1.00585	0.99390	-0.599	1.75×10^{-3}	5.70×10^{-4}
exterior cells/decade	40	460	281	1.00278	0.99082	-0.599	1.73×10^{-3}	3.67×10^{-4}
exterior cells/decade	80	920	281	1.00115	0.98920	-0.600	1.72×10^{-3}	2.79×10^{-4}
Y_{max}	10^2	380	281	1.00281	0.92734	-0.599	1.73×10^{-3}	3.66×10^{-4}
Y_{max}	10^3	420	281	1.00279	0.97276	-0.599	1.73×10^{-3}	3.66×10^{-4}
Y_{max}	10^4	460	281	1.00278	0.99082	-0.599	1.73×10^{-3}	3.67×10^{-4}
Y_{max}	10^5	500	281	1.00277	0.99802	-0.599	1.73×10^{-3}	3.67×10^{-4}
Y_{max}	10^6	540	281	1.00277	1.00088	-0.599	1.73×10^{-3}	3.66×10^{-4}
angular nodes	16	460	281	1.00331	0.99135	-0.595	1.72×10^{-3}	3.20×10^{-4}
angular nodes	32	460	281	1.00285	0.99090	-0.599	1.73×10^{-3}	3.60×10^{-4}
angular nodes	64	460	281	1.00278	0.99082	-0.599	1.73×10^{-3}	3.67×10^{-4}
angular nodes	128	460	281	1.00278	0.99083	-0.599	1.73×10^{-3}	3.66×10^{-4}
interior cells/decade	10	460	71	1.00357	0.99161	-0.599	4.22×10^{-3}	1.40×10^{-3}
interior cells/decade	20	460	141	1.00294	0.99098	-0.599	2.45×10^{-3}	5.58×10^{-4}
interior cells/decade	40	460	281	1.00278	0.99082	-0.599	1.73×10^{-3}	3.67×10^{-4}
interior cells/decade	80	460	561	1.00274	0.99078	-0.599	1.28×10^{-3}	3.23×10^{-4}

Table 9: Disk formula (7.11)–(7.12), analytic evaluation of Section 3.2: refinement of the exterior cells per decade, the truncation Y_{max} , the angular nodes and the interior cells per decade, one at a time about the base values (40, 10^4 , 64, 40). n_I is the number of interior cells; W_1 is measured against the base-level ring-mass CDF; the residual is that of (7.13) evaluated with the reconstructed ring masses.

ring-mass solve, the reconstruction and the radial particle dynamics agree in the W_1 sense to about 2×10^{-3} ; the boundary layer of the singular radial density is resolved only in the sense of mass allocation.

N	W_1	$\max_i \rho_i(T)$	Δt	W_1	$\max_i \rho_i(T)$	T	W_1	$\max_i \rho_i(T)$
90	2.91×10^{-3}	0.32391	0.100	2.64×10^{-3}	0.32391	50	2.63×10^{-3}	0.32391
180	2.63×10^{-3}	0.32391	0.050	2.63×10^{-3}	0.32391	100	2.89×10^{-3}	0.32358
360	2.47×10^{-3}	0.32391	0.025	2.63×10^{-3}	0.32392	150	3.05×10^{-3}	0.32351
720	2.41×10^{-3}	0.32391				200	3.15×10^{-3}	0.32351

Table 10: Radial particle system against the ring-mass CDF (W_1 distance): dependence on N ($\Delta t = 0.05$, $T = 50$; left), on Δt ($N = 180$, $T = 50$; middle) and on T ($N = 180$, $\Delta t = 0.05$; right).